

Ha Minh Hieu

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SUMMARY

Highly motivated Final-year Undergraduate and Computer Science Researcher at Hanoi University of Science and Technology, specializing in the intersection of Generative AI Applications, Optimization, and Autonomous Agentic AI. Primary research focuses on architecting collaborative Agentic Systems that orchestrate Generative AI to transform theoretical frameworks into robust, self-evolving solutions for accelerated optimization across complex, real-world decision-making domains, aiming to minimize manual intervention and maximize cost-efficiency within modern enterprise environments. Lead author and primary researcher distinguished by self-motivated research capabilities, successfully orchestrating collaborative synergy throughout the research lifecycle for multiple publications and manuscripts under peer review at premier A/A* conferences and Q1/Q2 journals, spanning AI applications, combinatorial optimization, AI for Scientific Discovery and healthcare, within elite academic and professional industrial settings.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL

Developer Tools: Git, VS Code, Visual Studio, PyCharm, Kaggle, Google Colab

AI & Machine Learning: LLMs, GNNs, Transformers, Agentic Design

Optimization: Evolutionary Computation, Metaheuristics, Combinatorial Optimization

Libraries: Pandas, NumPy, Matplotlib, Pytorch, Flask, Langchain, LangGraph, PyG

Other: Strong mathematical modeling and algorithm design background

EDUCATION

Hanoi University of Science and Technology

Hanoi, Vietnam

Talented Program – Computer Science

2022 – Present

- CPA: 3.64/4.0.

- Supervisor: Prof. Huynh Thi Thanh Binh

Lam Son High School for the Gifted

Thanh Hoa, Vietnam

Major in Mathematics

2019 – 2022

- CPA: 9.1/10

PUBLICATIONS

Conferences

- **Minh Hieu Ha**, Hung Phan, Tung Duy Doan, Tung Dao, Tran Cong Dao, Huynh Thi Thanh Binh. “Pareto-Grid-Guided Large Language Models for Fast and High-Quality Heuristics Design in Multi-Objective Combinatorial Optimization”, in AAAI Conference on Artificial Intelligence (AAAI) 2026, accepted as POSTER with 17.6%, (**rank A***).
- **Ha Minh Hieu**, Hung Phan, Duc Chinh Tran, Duc Cuong Van, Dao Van Tung, Huynh Thi Thanh Binh. “Alimentation Deep Multiple Optimal Ant Colony Optimization to solve Vehicle Routing Problem with Time Windows”, in GECCO '24 Companion: Proceedings of the Genetic and Evolutionary Computation Conference Companion, (**rank A**).

Journals

- Viet Tien Pham*, **Minh Hieu Ha***, Bao VQ Bui, Truong Son Hy. “FFTMd: leveraging fast-fourier transform for a lightweight and adversarial-resilient medical image segmentation framework”, in Scientific Reports, (**rank Q1**).
- Hanh Nguyen Thi, Cuong Van Due, Chinh Tran Due, **Hieu Ha Minh**, Son Nguyen Van, Quan La Van. “Memetic Algorithm for Maximizing K-coverage and K-Connectivity in Wireless Sensor Network”, in Informatica, (**rank Q2**).

Preprints

- **Minh Hieu Ha**, Khanh Ly Ta, Hung Phan, Tung Doan, Tung Dao, Dao Tran, Huynh Thi Thanh Binh. “MMP-A*: Multimodal Perception Enhanced Incremental Heuristic Search on Path Planning”, Under Review the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), (**rank A***).
- Tung Duy Doan, Tien Dat Vu, **Ha Minh Hieu**, Hung Phan, Tung Dao, Cong Dao Tran, Huynh Thi Thanh Binh. “TG-SR: Tree-Guided Search with Multiple Concept Priors for Symbolic Regression”, Under Review the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), (**rank A***).

WORK EXPERIENCE

Wing Lab

June 2026 – August 2026

Research Intern

NUS, Singapore

- Worked on AI for Scientific Discovery, with a focus on cost-efficient discovery using large language models.
- Developed **SpecEvo**: *Speculative Evolution with Large Language Models for Cost-Efficient Scientific Discovery*, a framework that generates effective code-based solutions at substantially lower LLM inference cost, inspired by speculative decoding and real-world scientific workflows.
- Supervisor: Prof. Min-Yen Kan.

VinBigData

July 2025 – February 2026

AI Engineer Intern

Ha Noi

- Hands-on training in machine learning, deep learning, and data analysis for real-world applications.
- Architected advanced RAG pipelines to significantly refine retrieval accuracy and mitigate hallucinations for complex user intents, without compromising system latency.
- Integrated long-term memory mechanisms to preserve cross-session user context, enabling the agent to deliver deeply personalized and proactive assistance within the VinFast ecosystem.

HySon Lab

May 2024 – July 2025

University of Alabama

Birmingham, United States

- Researched and developed foundation models in AI with applications spanning computer vision, biomedical analysis, and medical image segmentation.

Modeling, Simulation, and Optimization Laboratory

Sep 2023 – Present

BKAI, Hanoi University of Science and Technology (HUST)

Ha Noi

- Spearheaded the integration of Evolutionary Computation, Machine Learning, and Generative AI into large-scale optimization frameworks, specifically targeting autonomous robotic path planning, wireless sensor networks, and combinatorial optimization problems.
- Architected autonomous Agentic Systems that leverage LLM reasoning to automate complex algorithmic design and optimize mathematical functions, successfully transitioning theoretical paradigms into self-evolving solutions.
- Pioneered autonomous research into multi-agent reasoning and agentic design, focusing on the automated discovery of heuristics for real-world decision-making domains.
- Supervisor: Prof. Huynh Thi Thanh Binh.

AWARDS & HONORS

2026 Second Prize in the Scientific Research Student Award

Hanoi University of Science and Technology

2025 First Prize in Competition on LLM-designed Evolutionary Algorithms

The Genetic and Evolutionary Computation Conference (GECCO 2025) rank A

2025 First Runner-up in Competition on Numerical Global Optimization Competition on GNBG-II generated Test Suite

The Genetic and Evolutionary Computation Conference (GECCO 2025) rank A

2025 Second Prize in the Scientific Research Student Award

Hanoi University of Science and Technology

2025 Second Prize in AI4Industry Hackathon

Hanoi University of Industry

2024 Third Prize in the Scientific Research Student Award

Hanoi University of Science and Technology

2023 Third Prize in AI Challenge: ChatGPT Problem Solvers

Hanoi University of Science and Technology (HUST)

2022 Second Prize in Vietnamese Mathematical Olympiad

Ministry Of Education And Training Vietnam

2021 First Prize in Vietnamese Mathematical Olympiad

Ministry Of Education And Training Vietnam

LANGUAGE

- Vietnamese
- English

REFERENCES

Prof. Huynh Thi Thanh Binh

Professor, Vice Dean of the School of Information and Communication Technology (SoICT)

- Hanoi University of Science and Technology (HUST)
- Email: binhht@soict.hust.edu.vn
- Homepage: <https://users.soict.hust.edu.vn/binhht>